

# Heart Disease Prediction

CS-433 Project 1 – EPFL

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**Abstract**—Early prediction of cardiovascular diseases (CVD) can significantly reduce mortality. This task is complex due to noisy, imbalanced, and incomplete health data. In this project, we implement classical machine learning models from scratch to predict heart disease from the Behavioral Risk Factor Surveillance System (BRFSS)[1] data.

## I. INTRODUCTION

Cardiovascular diseases (CVD) are among the leading causes of death worldwide [2]. This project aims to build a binary classification model that predicts the presence of heart disease based on health and lifestyle data. The BRFSS dataset contains over 300,000 samples posing challenges such as missing values, inconsistent encodings, and class imbalance. To address these, we developed and compared regularized logistic regression models implemented entirely from scratch.

## II. MODELS AND METHODS

### A. Handling Missing Values and Data Imputation

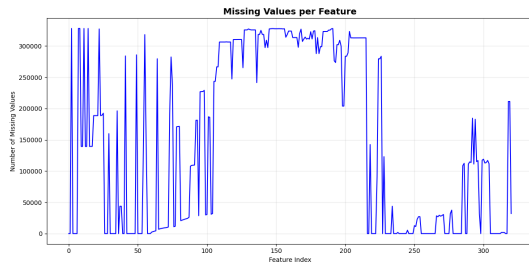


Fig. 1. Number of missing values per feature. The plot shows that several features contain a large amount of missing data, motivating the need for appropriate imputation strategies rather than feature removal.

We examined all 331 features to determine the most appropriate preprocessing strategy, including how to handle missing values (Figure 1). Features were divided into two types: categorical (one-hot encoded) and continuous (normalized). Missing values in categorical variables were imputed using a weighted random choice, while continuous features were imputed according to the distribution of valid responses.

For the `ALCDAY5` feature, representing the number of days in the past 30 during which the participant consumed alcohol, contained special codes: 777 (Don't know/Not sure), 888 (No drinks in past 30 days), and 999 (Refused to answer), along with missing entries (NaN).

The code 888 was treated as 0 drinking days. Codes 777, 999 and NaN were considered invalid and were imputed based on the empirical distribution of valid responses, ranging from 0 to maximum number of days. if 15% of participants reported

5 drinking days, then 15% of invalid or missing values were randomly replaced with 5. Even when many responses were missing, features were kept to preserve potentially informative variables.

A second important preprocessing step was the treatment of conditional or branching features, questions asked only if a previous answer met specific criteria. Each feature and its logic were manually reviewed to ensure consistent imputation.

As an example, in the case of the `ASTHNOW` feature (current asthma status) invalid codes (7, 9) were replaced using a weighted random choice based on valid responses (1 = Yes, 2 = No). Missing values were filled conditionally based on `ASTHMA3` (ever had asthma). If `ASTHMA3` = 2 (never had asthma), missing values in `ASTHNOW` were replaced with 3 (“not applicable”). For participants with `ASTHMA3` = 1, missing values were again imputed based on valid response distribution, ensuring internal consistency and preservation of the original data structure.

### B. One-Hot Encoding and Normalization

Categorical variables were transformed using one-hot encoding to convert discrete feature values into binary indicator variables. This allowed the models to interpret categorical information without assuming any ordinal relationship between categories.

Continuous variables were normalized to ensure all features contributed equally to the model training.

### C. Addressing Data Imbalance

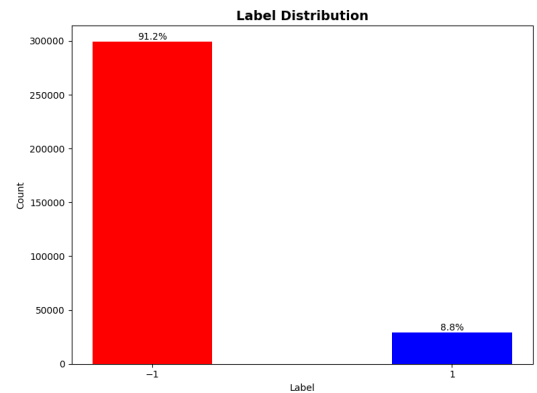


Fig. 2. Distribution of target labels in the dataset. The data are highly imbalanced, with approximately 91.5% of samples belonging to the negative class (no heart disease) and only 8.5% to the positive class (heart disease). This imbalance must be considered during model training and evaluation.

Figure 2 shows a strong class imbalance in the dataset. To mitigate this issue, we applied random oversampling of the minority class and undersampling of the majority class to achieve a more balanced distribution.

A balancing factor between 1 and 3 for undersampling and 0.5 and 1.5 for oversampling was used to control the class ratio. For instance, an oversampling factor of 0.5 increased the minority class to the half the size of the majority class, while an undersampling factor of 3 reduced the majority class to one third of the minority class. The optimal configuration, including the balancing factor and classification threshold, was determined by maximizing the mean F1 score under 5-fold cross-validation. This combination of rebalancing and threshold tuning improved the model’s sensitivity to positive (heart disease) cases.

#### D. Hyperparameter Tuning and Model Selection

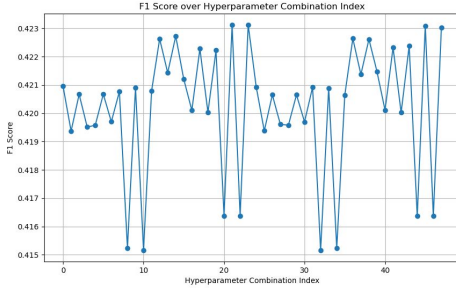


Fig. 3. F1 score across all hyperparameter combinations (one point per combination). Values shown correspond to the mean F1 score obtained through 5-fold cross-validation on the validation data; the highest value indicates the selected configuration.

We performed a grid search over model and preprocessing hyperparameters and selected the configuration maximizing the mean F1 score under 5-fold cross-validation. The search explored combinations of the following parameters:

- Regularization strength  $\lambda$
- Learning rate  $\gamma$
- Polynomial degree
- Regularization type  $\{L1, L2\}$
- Classification threshold
- Sampling strategy  $\{\text{undersampling, oversampling, none}\}$  with the corresponding sampling factor.

Figure 3 displays the F1 scores obtained for all hyperparameter combinations. After identifying the best region, a refined cross-validation was performed to fine-tune parameters. The final model, corresponding to the best refined parameters, was retrained on the full training set to obtain the optimal logistic regression weight vector  $w_{\text{optimal}}$  and generate the submission file for final evaluation.

Figure 4 illustrates how the sampling factor influenced the performance. The large standard deviation observed for lower factors indicates that the effect of sampling is strongly dependent on other hyperparameters changing one of them could significantly alter the optimal sampling factor.

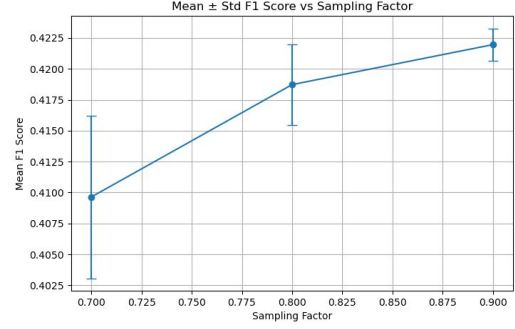


Fig. 4. Mean  $\pm$  standard deviation of F1 score as a function of the sampling factor. Each point represents the mean F1 score from 5-fold cross-validation for a given sampling factor, with error bars showing the standard deviation across folds.

#### E. Models Implemented

We implemented regularized logistic regression models from scratch, using both L1 (Lasso) and L2 (Ridge) penalties to prevent overfitting and improve generalization.

Given an input vector  $x_i \in R^d$  and label  $y_i \in \{-1, 1\}$ , the model predicts the probability of heart disease as:

$$P(y_i = 1|x_i) = \sigma(w^T x_i),$$

where  $\sigma(z) = \frac{1}{1+e^{-z}}$  is the sigmoid function and  $w$  the learned weight vector.

The cost function minimized during training was:

$$\mathcal{L}(w) = -\frac{1}{N} \sum_{i=1}^N \log(1 + e^{-y_i w^T x_i}) + \lambda R(w),$$

where  $R(w) = \|w\|_2^2$  for Ridge (L2) and  $R(w) = \|w\|_1$  for Lasso (L1) regularization.

Model parameters were optimized via gradient descent, with the learning rate  $\gamma$  and regularization strength  $\lambda$  tuned through cross-validation.

### III. RESULTS

Throughout sequential cross-validations, we identified the combination of hyperparameters and preprocessing strategies that maximized the model’s performance. After each cross-validation, the means  $\pm$  standard deviations of the test F1 scores were plotted (Figure 3 and Figure 4) with respect to the explored hyperparameters (see Section II-D), allowing us to iteratively refine the search space.

Although the hyperparameters were interdependent, this approach enabled our F1 scores to increase and observe certain global trends. Notably, oversampling produced the best results among sampling strategies (Figure 4) while the optimal decision threshold was found to be around 0.7 instead of the default 0.5. Finally, correlated features were not removed before model training because they consistently decreased our model evaluation metrics.

#### IV. DISCUSSION

Interestingly, the best results were achieved when only one feature was one-hot encoded, while all others were normalized. This suggests that some features initially treated as categorical were more informative when kept as ordinal variables.

Although manual imputation provided precise control over missing data, it was time-consuming and did not yield significantly better results.

#### V. SUMMARY

We developed and tuned regularized logistic regression models from scratch to predict heart disease using BRFSS data. Despite data imbalance and missing values, careful preprocessing and hyperparameter tuning led to steady performance improvements and valuable methodological insights.

#### REFERENCES

- [1] C. for Disease Control and Prevention, “Behavioral risk factor surveillance system (brfss),” [https://www.cdc.gov/brfss/annual\\_data/2015/pdf/codebook15\\_llcp.pdf](https://www.cdc.gov/brfss/annual_data/2015/pdf/codebook15_llcp.pdf), 2015, accessed: October 31, 2025.
- [2] W. H. Organization, “Cardiovascular diseases (cvds) fact sheet,” [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)), 2025, accessed: October 31, 2025.